Project Risk Management

SECTION: Definition

Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on at least one project objective (scope, schedule, cost, or quality). Note that risk can be an **opportunity** (positive) as well as a **threat** (negative) — good risk management captures both.

SECTION: The Risk Management Process

1. **Risk Management Planning** — deciding how risk activities will be conducted (methodology, roles, risk categories, tolerance levels).

2. **Risk Identification** — determining which risks might affect the project, documented in a **risk register**. Techniques include brainstorming, checklists, SWOT analysis, and expert interviews.

3. **Qualitative Risk Analysis** — assessing probability and impact of identified risks, usually on a scale (e.g. Low/Medium/High or 1–5), and plotting them on a **probability–impact matrix** to prioritize.

4. **Quantitative Risk Analysis** — numerically analyzing the effect of identified risks on overall project objectives (e.g. Monte Carlo simulation, decision tree analysis, Expected Monetary Value).

5. **Risk Response Planning** — developing options to reduce threats and enhance opportunities.

6. **Risk Monitoring and Control** — tracking identified risks, monitoring residual risks, identifying new risks, and evaluating the effectiveness of responses throughout the project.

SECTION: Risk Response Strategies

For threats (negative risks):

- **Avoid**: eliminate the threat by removing the cause (e.g. change the plan to avoid the risky activity).

- **Mitigate**: reduce the probability and/or impact to an acceptable threshold.

- **Transfer**: shift the impact to a third party (e.g. insurance, outsourcing, warranties) — this does not eliminate the risk, it moves ownership.

- **Accept**: acknowledge the risk without proactive action; can be *passive* (do nothing) or *active* (set aside a contingency reserve).

For opportunities (positive risks):

- **Exploit**: ensure the opportunity is realized (e.g. assign the best resources).

- **Enhance**: increase the probability or positive impact.

- **Share**: allocate ownership to a third party better able to capture the opportunity (e.g. a joint venture).

- **Accept**: take advantage if it happens, without actively pursuing it.

SECTION: Expected Monetary Value (EMV)

EMV = Probability × Impact (in monetary terms), summed across possible outcomes. It is used to compare risk-adjusted values of different response options or project paths, particularly in decision tree analysis.

Example: A risk has a 20% chance of causing a $50,000 cost overrun.

EMV = 0.20 × (–$50,000) = –$10,000 (this amount could be set aside as a contingency reserve).

SECTION: Contingency vs Management Reserve

- **Contingency Reserve**: budget/time allocated for *identified* risks ("known unknowns"); the PM can typically use this without additional approval.

- **Management Reserve**: budget/time held back for *unforeseen* risks ("unknown unknowns"); typically requires sponsor/management approval to access.

SECTION: Risk Register Contents

A risk register typically records: risk ID, description, category, probability, impact, risk score, response strategy, risk owner, and current status.

SECTION: Common Exam Angles

- Calculate EMV given probability and impact figures.

- Distinguish qualitative vs quantitative risk analysis.

- Match a scenario to the correct response strategy (avoid/mitigate/transfer/accept, or exploit/enhance/share/accept for opportunities).

- Explain the difference between contingency reserve and management reserve.